

Table 1: Final controlled $B_{c1} \rightarrow B_c \gamma$ comparison. Widths are in keV. The present result includes the explicitly integrated all-positive dimension-four G^2 sector. The contact/derivative rows have no ordinary two-channel double-Borel support. LHCb has observed the $B_c^+ \gamma$ structure, but the individual radiative partial widths are not measured.

Channel	Experiment	This work	Godfrey	Ebert et al.	Fulcher	Gershtein et al.	T. Y. Li et al.	Q. Li et al.	X. J. Li et al.	Eichten-Quigg	Bondar-Milstein
$B_{c1}(6743) \rightarrow B_c \gamma$	observed structure; partial width not measured	$0.532^{+0.202}_{-0.146}$	13	18.4	32.4	11.6	16.6	35	30.1	9.9	22
$B_{c1}(6750) \rightarrow B_c \gamma$	observed structure; partial width not measured	$10.04^{+2.41}_{-1.75}$	80	132	127.8	131.1	66.6	74	64	92.3	47

Table 2: Baseline $B_{c1} \rightarrow B_c^* \gamma$ comparison. Widths are in keV. The present vector result uses the hard-photon three-point term, the physical f_1, f_2 residues, and the explicitly integrated all-positive dimension-four G^2 sector. The vector hierarchy remains under the tensor-current sign and state-assignment audit. We use the self-contained standard vector-invariant normalization $f_{B_c^*} = 0.435^{+0.032}_{-0.035}$ GeV and the reference normalization $f_{B_c^*} = 0.384 \pm 0.038$ GeV only as a normalization cross-check. The contact-support audit finds no ordinary two-channel contact contribution.

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$B_{c1}(6743) \rightarrow B_c^* \gamma$	$70.3^{+10.9}_{-8.4}$ (standard $f_{B_c^*}$); $92.6^{+24.4}_{-16.6}$ (ref. $f_{B_c^*}$, perturbative)	60	78.9	75.6	77.8	49	70	47.8	62.5	75
$B_{c1}(6750) \rightarrow B_c^* \gamma$	382^{+55}_{-50} (standard $f_{B_c^*}$); 447^{+117}_{-83} (ref. $f_{B_c^*}$, perturbative)	11	13.6	26.2	8.1	10.5	40	25.6	7.5	2

Experimental status. The $B_c^+\gamma$ mass spectrum contains an observed $B_c(1P)^+$ -like structure, with a two-peak description around 6704.8 and 6752.4 MeV in the LHCb analysis. The partial radiative widths are not measured, so the experimental column should be read as an observation status rather than a numerical width comparison.

Interpretation. The quoted uncertainties are the Monte Carlo/input/Borel-window percentile intervals after folding in the explicitly integrated all-positive G^2 sector. The $B_c\gamma$ entries are the controlled predictions. The $B_c^*\gamma$ entries are convention-fixed mixed-current QCDSR values in the adopted two-point state assignment. Their high/low hierarchy is opposite to representative model calculations, but the independent commutator audit confirms the tensor-current sign; the reversal is traced to destructive interference in the 6743 amplitude and constructive interference in the 6750 amplitude. The separate contact audits leave no ordinary two-channel contact term to add to the standard double-Borel sum.